

# Ordinary abelian varieties over a finite field

Pierre Deligne (Bures-sur-Yvette)

We give a down-to-earth description of the category of ordinary abelian varieties over a finite field  $\mathbb{F}_q$ . The result obtained was inspired by Ihara [2], ch. V (see also [3]).

**1.** Let  $p$  be a prime number, let  $\mathbb{F}_p$  be the field  $\mathbb{Z}/(p)$ , let  $\overline{\mathbb{F}_p}$  be an algebraic closure of  $\mathbb{F}_p$ , and, for each power  $q$  of  $p$ , let  $\mathbb{F}_q$  be the subfield with  $q$  elements of  $\overline{\mathbb{F}_p}$ . For every algebraic extension  $k$  of  $\mathbb{F}_p$ , let  $W_0(k)$  denote the henselian discrete valuation ring, essentially of finite type over  $\mathbb{Z}$ , absolutely unramified, with residue field  $k$ ; let  $W(k)$  be the ring of Witt vectors over  $k$ , i.e. the completion of  $W_0(k)$ , let  $W = W(\overline{\mathbb{F}_p})$ , and let  $\varphi$  be an embedding of  $W$  in the field  $\mathbb{C}$  of complex numbers. We denote by  $\mathbb{Z}(1)$  the subgroup  $2\pi i\mathbb{Z}$  of  $\mathbb{C}$ . The exponential map defines an isomorphism between  $\mathbb{Z}(1) \otimes \mathbb{Z}_\ell$  and

$$\mathbb{Z}_\ell(1)(\mathbb{C}) = \varprojlim_m \mu_{\ell^m}(\mathbb{C}).$$

We denote by  $A^*$  the abelian variety dual to an abelian variety  $A$ . For every field  $k$ , let  $\bar{k}$  denote an algebraic closure of  $k$ .

**2.** Let  $A$  be an abelian variety of dimension  $g$ , defined over a field  $k$  of characteristic  $p$ . Recall that  $A$  is called ordinary if the following equivalent conditions are satisfied:

(I)  $A$  has  $p^g$  points of order dividing  $p$  with values in  $\bar{k}$ .

(II) The “Hasse–Witt matrix”

$$F^* : H^1(A^{(p)}, \mathcal{O}_{A^{(p)}}) \longrightarrow H^1(A, \mathcal{O}_A)$$

is invertible.

(III) The neutral component of the group scheme  $A_p$ , the kernel of multiplication by  $p$ , is of multiplicative type, hence geometrically isomorphic to a power of  $\mu_p$ .

If  $k = \mathbb{F}_q$ , if  $F$  is the Frobenius endomorphism of  $A$ , and if  $P_{cA}(F; x)$  is its characteristic polynomial, these conditions are further equivalent to:

(IV) At least half of the roots of  $P_{cA}(F; x)$  in  $\overline{\mathbb{Q}_p}$  are  $p$ -adic units. In other words, if  $n = \dim A$ , the reduction modulo  $p$  of the polynomial  $P_{cA}(F; x)$  is not divisible by  $x^{n+1}$ .

**3.** Let  $A$  be an ordinary abelian variety over  $\overline{\mathbb{F}_p}$ . Let  $\tilde{A}$  denote the canonical Serre–Tate lift [4] of  $A$  over  $W$ . Recall that  $\tilde{A}$  depends functorially on  $A$  and is characterized by the fact that the  $p$ -divisible group  $T_p(\tilde{A})$  over  $W$  attached to  $\tilde{A}$  [5] is the product of the  $p$ -divisible groups, uniquely determined by 2. (III), lifting respectively the neutral component and the largest étale quotient of  $T_p(A)$ . The canonical lift  $\tilde{A}$  is also the unique lift of  $A$  such that every endomorphism of  $A$  lifts to  $\tilde{A}$ . Let  $T(A)$  denote the integral homology of the complex abelian variety  $A_{\mathbb{C}}$  deduced from  $\tilde{A}$  and  $\varphi$  by extension of scalars from  $W$  to  $\mathbb{C}$ :

$$T(A) = H_1(\tilde{A} \otimes_{\varphi} \mathbb{C}).$$

It is known that  $\tilde{A}$  descends uniquely to  $W_0(\overline{\mathbb{F}_p})$ , so that  $A_{\mathbb{C}}$  depends only on  $A$  and on the restriction of  $\varphi$  to  $W_0(\overline{\mathbb{F}_p})$ . The free  $\mathbb{Z}$ -module  $T(A)$  has rank  $2 \dim(A)$ ; it is functorial in  $A$ . Moreover, if  $\ell$  is a prime number distinct from  $p$ , one has functorially

$$T(A) \otimes \mathbb{Z}_\ell = T_\ell(A). \quad (3.1)$$

The canonical lift of the abelian variety  $A^*$  dual to  $A$  is the dual of  $\tilde{A}$ , so that  $(A_{\mathbb{C}})^* = A_{\mathbb{C}}^*$  and  $T(A)$  and  $T(A^*)$  are in perfect duality with values in  $\mathbb{Z}(1)$ :

$$T(A) \otimes T(A^*) \longrightarrow \mathbb{Z}(1). \quad (3.2)$$

It is indispensable to use  $\mathbb{Z}(1)$  rather than  $\mathbb{Z}$  in order to obtain a theory invariant under complex conjugation. The pairings (3.2) are compatible, via (3.1), with the pairings

$$T_\ell(A) \otimes T_\ell(A^*) \longrightarrow \mathbb{Z}_\ell(1);$$

a morphism  $\xi : A \rightarrow A^*$  defines a polarization of  $A$  if and only if  $\xi_{\mathbb{C}} : A_{\mathbb{C}} \rightarrow A_{\mathbb{C}}^*$  defines a polarization of  $A_{\mathbb{C}}$ . Put

$$T'_p(A) = \text{Hom}(\mathbb{Q}_p/\mathbb{Z}_p, A(\overline{\mathbb{F}_p}))$$

and

$$T''_p(A) = \text{Hom}_{\mathbb{Z}_p}(T'_p(A^*), \mathbb{Z}(1) \otimes \mathbb{Z}_p).$$

These  $\mathbb{Z}_p$ -modules are covariant functors of  $A$ .

By definition of the canonical lift, the  $p$ -divisible group  $T_p(\tilde{A})$  is the sum of the constant pro-étale group  $T'_p(A)$  and of the Cartier dual of  $T'_p(A^*)$ . For every morphism  $u : A \rightarrow B$ , the induced morphism

$$u : T_p(\tilde{A}) \rightarrow T_p(\tilde{B})$$

is identified with the sum of  $u|_{T'_p(A)} : T'_p(A) \rightarrow T'_p(B)$  and the Cartier transpose of  ${}^t u|_{T'_p(B^*)} : T'_p(B^*) \rightarrow T'_p(A^*)$ . Over  $\mathbb{C}$ , one has canonically  $\mathbb{Z}(1)/(p^n) \sim \mu_{p^n}$ , whence an isomorphism of functors:

$$T_{(p)}(A) = T(A) \otimes \mathbb{Z}_p = T'_p(A) \oplus T''_p(A). \quad (3.2)$$

**4.** Recall that, if  $\varphi : X \rightarrow Y$  is an isogeny between complex abelian varieties, the exact homotopy sequence reduces to a short exact sequence

$$0 \longrightarrow H_1(X) \longrightarrow H_1(Y) \longrightarrow \text{Ker}(\varphi) \longrightarrow 0.$$

The abelian varieties which are quotients of  $X$  by a finite subgroup, and these finite subgroups of  $X$ , correspond bijectively to the sublattices of  $H_1(X) \otimes \mathbb{Q}$  containing  $H_1(X)$ .

Let  $A$  be an ordinary abelian variety over  $\overline{\mathbb{F}_p}$ . If  $n$  is an integer prime to  $p$ , the finite group subschemes of order  $n$  of  $A$ , of  $\tilde{A}$ , and of  $A_{\mathbb{C}}$  correspond bijectively, and still correspond to the overlattices  $R$  of  $T(A)$  such that  $[R : T(A)] = n$ .

Put

$$V'_p = T'_p(A) \otimes \mathbb{Q}_p, \quad V''_p(A) = T''_p(A) \otimes \mathbb{Q}_p.$$

The finite group subschemes of order  $p^k$  of  $A$  are products of an étale subgroup and an infinitesimal subgroup. The étale subgroups of order  $p^k$  of  $A$  correspond to those subgroups of order  $p^k$  of  $A_{\mathbb{C}}$  for which the corresponding overlattice  $R$  of  $T(A)$  is contained in  $T_{(p)}(A) + V'_p(A)$ . By duality, the infinitesimal subgroups of  $A$  correspond to the overlattices  $R$  of  $T(A)$ ,  $p$ -isogenous to  $T(A)$ , i.e. such that  $[R : T(A)]$  is a power of  $p$ , and contained in  $T_{(p)}(A) + V''_p(A)$ .

In all, the finite subgroups of  $A$ , or the abelian varieties quotient of  $A$ , correspond bijectively to the overlattices  $R$  of  $T(A)$  such that

$$R \otimes \mathbb{Z}_p = (R \otimes \mathbb{Z}_p \cap V'_p) + (R \otimes \mathbb{Z}_p \cap V''_p). \quad (4.1)$$

**5.** In particular,  $A^{(p)}$ , the quotient of  $A$  by the largest infinitesimal subgroup of  $A$  killed by  $p$  for  $A$  ordinary, is defined by the overlattice  $T(A)^{(p)}$  of  $T(A)$ ,  $p$ -isogenous to  $T(A)$ , such that

$$T(A)^{(p)} \otimes \mathbb{Z}_p = T'_p(A) + \frac{1}{p}T''_p(A).$$

**6.** Let  $A$  be an abelian variety over  $\mathbb{F}_q$  and let  $F : x \mapsto x^q$  be its Frobenius endomorphism. Recall that  $A$  is uniquely determined by the pair  $(\bar{A}, F)$  deduced from  $(A, F)$  by extension of scalars from  $\mathbb{F}_q$  to  $\overline{\mathbb{F}_q}$ ; the endomorphism  $F$  of  $\bar{A}$  factors through the relative Frobenius

$$\text{Fr}^{(q)} : \bar{A} \longrightarrow \bar{A}^{(q)}$$

and an isomorphism

$$F' : \overline{A}^{(q)} \longrightarrow \overline{A}.$$

If  $A$  is ordinary,  $T(A)$  will denote the  $\mathbb{Z}$ -module  $T(\overline{A})$  endowed with the endomorphism  $F$  induced by the Frobenius endomorphism of  $A$ . By 5, by what precedes, and by (3.2), the lattices  $T(A)$  and  $F(T(A))$  are  $p$ -isogenous and one has

$$F(T'_p(A)) = T'_p(A), \quad (6.1)$$

$$F(T''_p(A)) = qT''_p(A). \quad (6.2)$$

**7. Theorem.** The functor

$$A \longmapsto (T(A), F)$$

is an equivalence of categories between the category of ordinary abelian varieties over  $\mathbb{F}_q$  and the category of finite-type free  $\mathbb{Z}$ -modules  $T$  endowed with an endomorphism  $F$  satisfying the following conditions:

- (a)  $F$  is semisimple, and its eigenvalues have complex absolute value  $q^{1/2}$ .
- (b) At least half of the roots in  $\overline{\mathbb{Q}}_p$  of the characteristic polynomial of  $F$  are  $p$ -adic units; in other words, if  $T$  has rank  $d$ , the reduction modulo  $p$  of the polynomial  $P_{cT}(F; x)$  is not divisible by  $x^{\lfloor d/2 \rfloor + 1}$ .
- (c) There exists an endomorphism  $V$  of  $T$  such that  $FV = q$ .

If condition (a) is satisfied, conditions (b) and (c) are equivalent to:

- (d) The module  $T \otimes \mathbb{Z}_p$  admits a decomposition, stable under  $F$ , into two  $\mathbb{Z}_p$ -submodules  $T'_p$  and  $T''_p$  of the same dimension, such that  $F|_{T'_p}$  is invertible and  $F|_{T''_p}$  is divisible by  $q$ .

*Proof.* (A) We prove that (a)+(b)+(c) imply (d). If  $\alpha$  is a complex eigenvalue of  $F$ , then  $\bar{\alpha}$  is another, with the same multiplicity, and  $\alpha\bar{\alpha} = q$ . Apart from those equal to  $\pm q^{1/2}$ , the eigenvalues of  $F$  in  $\mathbb{C}$ , hence in  $\overline{\mathbb{Q}}_p$ , fall into pairs of roots  $\alpha$  and  $q/\alpha$ . The roots  $\alpha$  and  $q/\alpha$  cannot both be  $p$ -adic units; it follows from (b) that  $\pm q^{1/2}$  is not an eigenvalue of  $F$ , that half of the eigenvalues of  $F$  in  $\overline{\mathbb{Q}}_p$  are  $p$ -adic units, say  $\alpha_1, \dots, \alpha_{d/2}$ , and that the other half consists of  $\beta_1 = q/\alpha_1, \dots, \beta_{d/2} = q/\alpha_{d/2}$ . Let

$$T_{(p)} = T \otimes \mathbb{Z}_p, \quad V_p = T \otimes \mathbb{Q}_p,$$

let  $V'_p$  be the subspace of  $V_p$  which is the kernel of  $\prod_i (F - \alpha_i)$ , and let  $V''_p$  be the kernel of the endomorphism  $\varphi = \prod_i (F - \beta_i)$ . One has

$$V_p = V'_p \oplus V''_p.$$

Let  $T'_p$  be the projection of  $T_{(p)}$  onto  $V'_p$  and let  $T''_p = T_{(p)} \cap V''_p$ . Since  $\varphi$  kills  $V''_p$  and preserves  $T$ , it sends  $T'_p$  into  $T_{(p)} \cap V'_p \subset T'_p$ . On the other hand,

$$\det(\varphi|_{V'_p}) = \prod_{i,j} (\alpha_i - \beta_j)$$

is a  $p$ -adic unit, hence  $\varphi(T'_p) = T'_p$  and  $T_{(p)} \cap V'_p = T'_p$ , so that

$$T_{(p)} = T'_p \oplus T''_p.$$

(B) *Full faithfulness.* Let  $A$  and  $B$  be two abelian varieties over  $\mathbb{F}_q$ , and let  $\psi$  be the map

$$\psi : \text{Hom}(A, B) \longrightarrow \text{Hom}_F(T(A), T(B)).$$

By Tate's theorem [7] and (3.1), the map

$$\psi_\ell : \text{Hom}(A, B) \otimes \mathbb{Z}_\ell \longrightarrow \text{Hom}_F(T(A), T(B)) \otimes \mathbb{Z}_\ell$$

is an isomorphism for  $(\ell, p) = 1$ , so that  $\psi \otimes \mathbb{Q}$  is an isomorphism. It is known that  $\text{Hom}(A, B)$  is torsion-free, hence  $\psi$  is injective. Let  $u : A \rightarrow B$  be a morphism such that  $T(u)$  is divisible by  $n$ . The induced morphism

$$u_{\mathbb{C}} : A_{\mathbb{C}} \longrightarrow B_{\mathbb{C}}$$

is then divisible by  $n$ , hence so is  $\tilde{u} : \tilde{A} \rightarrow \tilde{B}$  at the generic point of  $W$ . The kernel of multiplication by  $n$  is flat over  $W$ ; thus  $u$  vanishes on this kernel,  $\tilde{u}$  and  $u$  are divisible by  $n$ , and  $\psi$  is bijective.

(C) *Necessity.* That  $(T(A), F)$  satisfies (a) follows from Weil; condition (d), which implies (b) and (c), follows from 6.

(D) *Isogenies.* Let  $(T_0, F)$  satisfy (a) (d) and let  $T$  be a lattice in  $T_0 \otimes \mathbb{Q}$ , stable under  $F$ , still satisfying (d). Suppose that  $(T_0, F)$  is the image of an abelian variety  $A$  over  $\mathbb{F}_q$  and let us prove that  $(T, F)$  comes from an isogenous abelian variety. Replacing  $T$  by  $\frac{1}{k}T$ , which is isomorphic to it, one may suppose that  $T \supset T_0$ . Condition (d) implies that  $T$  satisfies (4.1), and  $T$  defines a subgroup  $H$  of  $\bar{A}$ , which is defined over  $\mathbb{F}_q$ , and

$$(T, F) = T(A/H).$$

(E) *Surjectivity.* The functor  $T$  induces a functor  $T_{\mathbb{Q}}$  from the category of ordinary abelian varieties up to isogeny over  $\mathbb{F}_q$  to the category of finite-rank  $\mathbb{Q}$ -vector spaces endowed with an automorphism  $F$  satisfying (a) (b). By (D), it is enough to prove that this functor  $T_{\mathbb{Q}}$  is essentially surjective. It is even enough to show that every simple object  $(V, F)$  of the image category is attained. By Honda [1] (see also [6]), there exists an abelian variety  $A$  over  $\mathbb{F}_q$  such that the characteristic polynomial of the Frobenius  $F_A$  of  $A$  is a power of that of  $F$ . The third characterization in 1. of ordinary abelian varieties shows that  $A$  is ordinary. Moreover,  $(T(A) \otimes \mathbb{Q}, F)$  is a sum of copies of  $(V, F)$ , hence, by (B), the abelian variety up to isogeny  $A \otimes \mathbb{Q}$  is a sum of copies of an abelian variety  $B$  satisfying

$$T(B) \otimes \mathbb{Q} = (V, F).$$

8. Let  $(T, F)$  be a pair satisfying the hypotheses of the theorem, let  $2g$  be the rank of  $T$ , let  $A$  be the corresponding abelian variety over  $\mathbb{F}_q$ , and let  $A_{\mathbb{C}}$  be the complex abelian variety deduced from it (3.). One has

$$T = H_1(A_{\mathbb{C}}),$$

so that  $T \otimes \mathbb{R}$  is identified with the Lie algebra of  $A_{\mathbb{C}}$  and, as such, is endowed with a complex structure. Here is, according to J.-P. Serre, how to reconstruct this complex structure in terms of  $T$ ,  $F$ , and the restriction of  $\varphi$  to  $W_0(\overline{\mathbb{F}_p})$ .

**Proposition.** The complex structure defined above on  $T \otimes \mathbb{R}$  is characterized by the following properties:

- (I) The endomorphism  $F$  is  $\mathbb{C}$ -linear.
- (II) If  $v$  is the valuation of the algebraic closure  $\overline{\mathbb{Q}}$  of  $\mathbb{Q}$  in  $\mathbb{C}$  extending the valuation of  $W_0(\overline{\mathbb{F}_p})$ , the valuations of the  $g$  eigenvalues of this endomorphism are strictly positive.

Condition (I) is evident, and condition (II) follows from the fact that the action of  $F$  on the Lie algebra of  $A$  is congruent to zero modulo  $p$ . The uniqueness of the complex structure satisfying (I) and (II) follows easily from condition (b) satisfied by  $(T, F)$ .

### Bibliography

1. Honda, T.: Isogeny classes of abelian varieties over finite fields. J. Math. Soc. Jap. **20**, 83–95 (1968).
2. Ihara, Y.: On congruence monodromy problems, vol. I. University of Tokyo, 1968.
3. — The congruence monodromy problems. J. Math. Soc. Jap. **20**, 107–121 (1968).
4. Lubin, J., J.-P. Serre, and J. Tate: Elliptic curves and formal groups. Woods Hole Summer Institute 1964 (mimeographed, printed in a restricted number of copies).
5. Serre, J.-P.: Groupes  $p$ -divisibles (d’après J. Tate). Séminaire Bourbaki 318, Novembre 1966.
6. Tate, J.: Classes d’isogénies de variétés abéliennes sur un corps fini (d’après T. Honda). Séminaire Bourbaki 352, Novembre 1968.
7. — Endomorphisms of abelian varieties over finite fields. Inventiones Math. **2**, 134–144 (1966).

Pierre Deligne  
I.H.E.S.  
91, Bures-sur-Yvette, France

(Received July 3, 1969)